Outline Draft

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1 Introduction

Alzheimer's Disease (AD) is a complex neurodegenerative disease affecting more than 55 million people globally [14]. Symptoms are caused by the degeneration of the central nervous system, and are manifested by progressive memory loss, difficulty with language, drastic mood swings, among other effects. It is a disease that causes an extensive burden to those suffering from the disease through disorientation, to relatives in the form of mental taxation, as well as economically, projecting an estimated global economic burden of 16.9 trillion dollars by 2050 [18]. These factors are just a few reasons as to the importance of research in this area, where many scientists have worked to understand the underlying mechanisms of AD. However, much is still unknown in how AD is caused, as well as what biological processes occur in the brain, making it difficult to come to a decisive conclusion regarding its cause.

Many studies have been conducted in regard to understanding Alzheimer's Disease. With this, there are a few main hypotheses that are generally considered to be factors in the causation and progression of AD [22]. The *Amyloid Cascade Hypothesis* is among the most extensively studied in the literature, where it is proposed that a misfolding of β-amyloid (A_β) protein forms senile plaques. These plaques are hypothesized to be the main driving force in the intracellular deposition of misfolded τ proteins in neurofibrillary tangles (NFTs) and memory loss in the progression of Alzheimer's Disease [9]. Growing evidence, however, shows that this hypothesis is suggested to be but a small factor in the larger picture of the underlying operative mechanisms of the disease [12]. This has led researchers to branch into other pathways concerning AD, looking at the interactions between A_β and τ in an attempt to further understand the pathogenesis of these two factors [21]. Some studies have also observed the role of calcium in this process, further suggesting a more complex interaction that is occurring biologically. This occurs through a feedback loop between an upregulation of Ca^{2+} and the amyloidogenic pathway, remodeling the Ca^{2+} signaling pathway [4].

The various complex interactions in AD have shown to be large obstacles when modeling the behavior of the disease. Numerous different studies have in turn made an attempt at creating mathematical models observing different aspects of the disease in order to analyze the underlying dynamics. This has led to many models being introduced, each aiming to understand various components in Alzheimer's Disease through different methodologies. Compartments concerning the interaction between A_β and τ are studied, observing the suspected effects on neurodegeneration and cognitive decline through a mechanistic causal model [20, 8]. The feedback loop between A_β and Ca^{2+} has also been analyzed using a deterministic system, adding a noise-inducing element to the system [19, 23]. Models that observe lower-level processes have also been studied, lending to valuable insight as to what is seen on the cellular level [15, 13].

It should be noted that in these models, while each has its own way of observing some of the underlying mechanisms, there are still potential shortfalls when it comes to finding effective treatments. Evidence suggests that focusing solely on anti-A_β treatments is ineffective [1, 21, 24]. This provides difficulty in understanding exactly the importance of which component of the disease should be the focus of further study in order to provide evidence for treatments having higher efficacy. In this study, a mechanistic model is proposed, analyzing the interaction between A_β, Ca^{2+}, and τ proteins. We will observe the effects of this multifaceted interaction on neurodegeneration and cognitive decline. Sensitivity analysis is performed in order to better understand the importance of each aspect of the model, and optimal control is used in order to determine the suggested avenues of effective treatment.

2 Proposed Model

To understand the role that calcium may have in the development of AD, a model is proposed using three biomarkers found to be substantial during disease progression: beta-amyloid and tau, as well as the interactional biomarker calcium, which is suggested to have a role in beta-amyloid and tau protein accumulation. Also observed are neuronal loss and cognitive decline. The pathology of AD is suggested to start through a feedback loop between beta-amyloid and calcium through a dysregulation of calcium, which then exacerbates the aggregation of beta-amyloid, forming plaques. These plaques then promote the hyperphosphorylation of tau proteins, leading to neurodegeneration and subsequently cognitive decline. The model will describe the progression of Alzheimer's disease from a certain age; we use the arbitrary starting age of 50 in our model.

2.1 Amyloid Beta and Calcium Equations

Because amyloid beta and calcium are suggested to interact in a way that a dysregulation of calcium causes an acceleration of amyloid beta accumulation [19], this is modeled mathematically as

$$\frac{dA_\beta}{dt} = a_1 + a_2\frac{Ca^n}{Ca^n + \sigma_1^n} - k_1 A_\beta - u_1 A_\beta \quad \text{with} \quad A_\beta(T_0) = A_0, \tag{1}$$

where A_0 is the initial condition of A_β at age T_0, and can vary on a per patient basis [8]. The rate of A_β growth increases at a linear rate a_1 and decreases proportionally to the population of A_β, as described by $k_1 A_\beta$. Here, $a_2\frac{Ca^n}{Ca^n+\sigma_1^n}$ acts as the interaction of calcium on A_β using a Hill equation, where we choose $n = 1$. In previous literature, Hill equations of $n = 2$ are implemented in order to have bistability and (other stuff) (citation needed). Since our model only seeks to describe patients that already have Alzheimer's Disease, we use $n = 1$, which also provides the benefit of tractable stability solutions. The removal rate can be modified by choosing different values for u_1 which represents the effect of treatment targeting A_β. The term u_1 can either be constant or be a function of time.

Similarly, an A_β interaction is observed to have a role in dysregulation of calcium, and the equation for calcium is defined as

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \quad \text{with} \quad Ca(T_0) = Ca_0. \tag{2}$$

Here, calcium grows at a linear rate b_1, and the accumulation due to A_β is positive at a rate b_2. There is also a natural removal rate of calcium, k_2. Notice that there is also the option to administer treatment targeting calcium through the term u_2.

2.2 Tau Equation

Many studies have suggested that the pathology of tau is initiated by amyloid beta accumulation, and have modeled this process mathematically [8, 25]. However, it has also been suggested that calcium dysregulation has an effect inducing the production of tau proteins [7]. Since no calcium-tau interaction is found in dynamical systems literature, we implement a Hill equation similar to that of Equation (1). Thus, the tau equation is given by,

$$\frac{d\tau}{dt} = c_1 + c_2 A_\beta + c_3 \frac{Ca^n}{Ca^n + \sigma_2^n} - k_3\tau - u_3\tau \quad \text{with} \quad \tau(T_0) = \tau_0. \tag{3}$$

It is noted here that only pathological tau is observed, where other studies have used compartments relating to non-amyloid dependent tau accumulation, or suspected non-Alzheimer pathology (SNAP) [8, 20]. This function is similar to Equation (1) and Equation (2) in that $\frac{d\tau}{dt}$ grows at a linear rate d_1 and decreases proportional to the population of τ through $k_3\tau$. There is also the treatment term $u_3\tau$. Again, we choose $n = 1$ for tractability of stability solutions and further analysis.

2.3 Neurodegeneration Equation

When tau is in a pathological state, it no longer binds to microtubules as intended; instead, the formation of NFTs occur in the neurons, leading to neurodegeneration [2]. Hence, this equation is defined as

$$\frac{dN}{dt} = d_1 + d_2\tau - k_4 N.$$

This equation follows the established structure and also includes the interaction term $d_2\tau$. There are no treatment terms for the N and C equations.

2.4 Cognitive Decline Equation

Both tau and neurodegeneration are shown to directly affect cognitive decline [3, 8]. Because of this, the equation for cognitive decline is defined as

$$\frac{dC}{dt} = e_1 + e_2 NR + e_3\tau - k_5 C.$$

The parameter R represents a patient having the APOE $\epsilon 4$ gene, which has been shown to be a risk factor in Alzheimer's disease [20].

This model is summarized as the following system of ordinary differential equations:

$$
\begin{aligned}
\frac{dA_\beta}{dt} &= a_1 + a_2\frac{Ca}{Ca+\sigma_1} - k_1A_\beta - u_1A_\beta \\
\frac{dCa}{dt} &= b_1 + b_2A_\beta - k_2Ca - u_2Ca \\
\frac{d\tau}{dt} &= c_1 + c_2A_\beta + c_3\frac{Ca}{Ca+\sigma_2} - k_3\tau - u_3\tau \\
\frac{dN}{dt} &= d_1 + d_2\tau - k_4N \\
\frac{dC}{dt} &= e_1 + e_2NR + e_3\tau - k_5C
\end{aligned}
\tag{4}
$$

with the initial conditions:

$$
\begin{cases}
A_\beta(T_0) = A_0 \\
Ca(T_0) = Ca_0 \\
\tau(T_0) = \tau_0 \\
N(T_0) = N_0 \\
C(T_0) = C_0.
\end{cases}
$$

3 Analysis

3.1 Preliminary Results

Assuming system (4) is subject to non-negative initial conditions, the system is invariant in the non-negative orthant. Let $r_n = k_n + u_n$ for the simplicity of calculations. Considering the components uncoupled from the rest of the proposed model, dA_β/dt and dCa/dt can be shown to be bounded. First, notice that

$$
a_1 + a_2\frac{Ca}{Ca+\sigma_1} - r_1A_\beta \leq a_1 + a_2 - r_1A_\beta
$$

since $Ca/(Ca+\sigma_1) \in [0,1)$. So, $a_1 + a_2 - r_1A_\beta < 0 \implies dA_\beta/dt < 0$ when $A_\beta > \frac{a_1+a_2}{r_1}$. Similarly, using the fact A_β is bounded, it follows that

$$
\frac{dCa}{dt} = b_1 + b_2A_\beta - r_2Ca \leq b_1 + b_2M - r_2Ca
$$

as $M > A_\beta(t)$ for all $t \geq 0$. So, $b_1 + b_2M - r_2Ca < 0 \implies dCa/dt < 0$ when $Ca > \frac{b_1+b_2M}{r_2}$, thus Ca is also bounded. The rest of the system follows from these two variables, so the rest of the system is bounded as well.

Lemma 1 (Establishment of Unique Equilibria of Uncoupled System)**.** *There exists a unique equilibrium (A_β^*, Ca^*) for the system (A_β, Ca).*

Proof. To establish the existence of equilibria for the system, a solution to the following system of equations must be obtained:

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2\frac{Ca}{Ca+\sigma_1} - r_1 A_\beta = 0 \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca = 0. \end{cases}$$

By solving dA_β/dt for A_β, we get

$$A_\beta = \frac{a_1 Ca + a_2 Ca + a_1\sigma_1}{r_1(Ca+\sigma_1)}.$$

We then substitute this value into dCa/dt. The resulting expression is

$$b_1 - r_2 Ca + \frac{b_2(a_1 Ca + a_2 Ca + a_1\sigma_1)}{r_1(Ca+\sigma_1)}. \tag{5}$$

Setting Equation (5) equal to 0 and doing further simplification yields the quadratic,

$$-(r_1 r_2)Ca^2 + (a_1 b_2 + a_2 b_2 + b_1 r_1 - r_1 r_2 \sigma_1)Ca + a_1 b_2 \sigma_1 + b_1 r_1 \sigma_1 = 0. \tag{6}$$

Notice that in Equation (6) the quadratic term is always negative and the constant term is always positive. The linear term can either be positive or negative depending on the choice of parameters. Regardless, there is only one sign change in the expression, so by Descartes' Rule of Signs, there is one positive real root to this quadratic. Thus, there exists a unique positive equilibrium (A_β^*, Ca^*) for the submodel. □

Lemma 2 (Global Stability of Equilibrium)**.** *The equilibrium (A_β^*, Ca^*) is globally asymptotically stable.*

Proof. Let $\mathbf{F}(A_\beta, Ca) = \left\langle a_1 + \frac{a_2 Ca}{Ca+\sigma_1} - r_1 A_\beta,\, b_1 + b_2 A_\beta - r_2 Ca \right\rangle$. Then, $\nabla \cdot \mathbf{F} = -r_1 - r_2$. Choose the function $\alpha(A_\beta, Ca) = 1$. So, $\nabla\cdot(\alpha\mathbf{F}) = -r_1 - r_2 < 0$, therefore by the Bendixson-Dulac Theorem, there is no limit cycle in the system of ODEs. So, the equilibrium point in the non-negative orthant must be globally asymptotically stable. □

Theorem 1 (Global Stability of Equilibrium of System)**.** *There exists a unique globally asymptotically stable equilibrium $(A_\beta^*, Ca^*, \tau^*, N^*, C^*)$.*

Proof. By Lemma 2, there is a unique globally asymptotically stable equilibrium (A_β^*, Ca^*) for the (A_β, Ca) submodel. By solving (3) for τ we get the following equilibrium value:

$$\tau^* = \frac{c_3 Ca^* + c_1(Ca^* + \sigma_2) + A_\beta^* c_2(Ca^* + \sigma_2)}{r_3(Ca^* + \sigma_2)}$$

By similar methods we find

$$N^* = \frac{d_1 + d_2\tau^*}{k_4}$$

and

$$C^* = \frac{e_1 + e_2 N^* R + e_3\tau^*}{k_5}.$$

Thus there is a unique global asymptotically stable equilibrium solution for system (4). □

4 Parameters

The values for the 19 key parameters in the causal model are obtained and estimated from the literature, and are summarized in Table 1. In order to provide a plausible model, parameters used in previous studies are adjusted to allow for similar units and orders of magnitude. When this is not possible, parameters are estimated using values found from various biological studies as are credible.

Table 1: Parameters used for simulation

Var	Value	Unit	Source
A_0	0	micromolars (µM)	-
Ca_0	100	nanomolars (nM)	[16]
τ_0	0	µM	-
N_0	0	billions of neurons lost	-
C_0	0	MMSE test score decline	-
a_1	6.5641	$\mu M \cdot year^{-1}$	[15] adjusted
a_2	1.5778	$\mu M \cdot year^{-1}$	[23] adjusted
b_1	315569260	$nM \cdot year^{-1}$	[23] lets review this one
b_2	6311385.2	$year^{-1}$ (100)	[23]
c_1	52.2958	$\mu M \cdot year^{-1}$	[17]
c_2	1.78367	$year^{-1}$	[15]
c_3	0.1	$year^{-1}$ (1/100)	no source
d_1	0.0716	billions of neurons lost $\cdot$ $year^{-1}$	no source
d_2	0.398406	billions of neurons lost $\cdot$ $year^{-1} \cdot \mu M^{-1}$	[17]
e_1	0.01463	MMSE decline $\cdot$ $year^{-1}$	no source
e_2	0.008684	MMSE decline $\cdot$ $year^{-1}$ $\cdot$ billions neurons $lost^{-1}$	[8] adjusted
e_3	0.01992	MMSE decline $\cdot$ $year^{-1} \cdot \mu M^{-1}$	[8] adjusted
k_1	0.303598	$year^{-1}$	[15] adjusted
k_2	3155692.6	$year^{-1}$	[23]
k_3	1.8333	$year^{-1}$	no source
k_4	0.3588	$year^{-1}$	no source (?)
k_5	0.08684	$year^{-1}$	no source
u_1	varies	$year^{-1}$	-
u_2	varies	$year^{-1}$	-
u_3	varies	$year^{-1}$	-
σ_1	$50 - 150$	nM	no source (?)
σ_2	$50 - 150$	nM	no source (?)

5 Sensitivity Analysis

Sobol sensitivity analysis is used to provide a quantitative assessment of the impact that model parameters have on the causal dynamics of the proposed model. This computation was performed using Python with the SALib library [10, 11]. With a final time $t = 50$ years after age 50, the first and total-order indices (S_T) were computed using different sample sizes, ranging from $N = 2^4$ to $N = 2^{20}$. Using the parameter ranges from Table 1, the first-order and total-order sensitivity indices are highlighted in Figure 1; initial conditions were set for $Ca(0) = 100$, and zero for the rest of the state variables. In order to validate these indices, Figure 2 shows the six most sensitive parameters for each sample size. The total-order indices are quantified along with the confidence interval for each parameter at $N = 2^{16}$ in Table 2. It is noted that the confidence intervals for each parameter being less than 10% of S_T indicates a high level of accuracy. Using a relative cutoff, the threshold is calculated by Threshold $= 0.01 \cdot \max(S_{T_i})$. This determines what parameters are negligible, relative to the highest scoring sensitivity indices. It is observed that c_3 is negligible. This parameter being less than the threshold suggests that, though calcium interplays with tau in the process of Alzheimer's Disease, it may be a minor factor in terms of its interaction on the hyperphosphorylation of tau.

Table 2: Total Sobol Sensitivity indices for various parameters with corresponding confidence intervals for $N = 2^{20}$

Param	k_3	k_5	e_3	c_1	k_1	c_2	a_1
ST	2.783e-01	2.432e-01	1.244e-01	8.047e-02	6.020e-02	5.824e-02	4.607e-02
Conf Int	7.929e-04	7.580e-04	3.946e-04	2.732e-04	1.891e-04	1.871e-04	1.506e-04

Param	k_4	r	e_2	d_2	a_2	k_2	σ_1
ST	3.021e-02	3.013e-02	3.013e-02	2.994e-02	9.639e-04	1.577e-04	1.544e-04
Conf Int	9.158e-05	9.651e-05	9.844e-05	9.105e-05	3.222e-06	4.884e-07	4.757e-07

Param	b_1	e_1	b_2	d_1	c_3	σ_2
ST	7.116e-05	2.352e-05	1.760e-05	3.409e-07	1.064e-07	1.720e-08
Conf Int	2.544e-07	7.665e-08	6.758e-08	1.135e-09	3.273e-10	7.593e-11

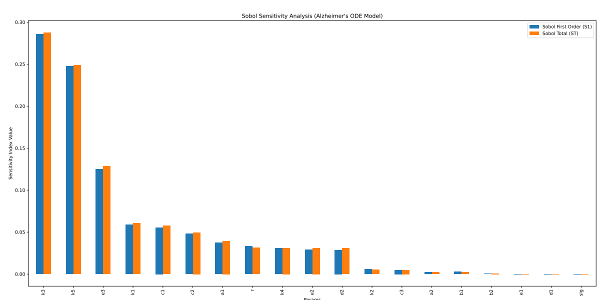

Figure 1: First and Total order indices at $N = 2^{10}$

6 Results

Using the derived biological parameters in Table 1, we run the simulation forward for 50 years with the initial conditions $(A_0, Ca_0, \tau_0, N_0, C_0) = (0, 100, 0, 0, 0)$. This simulation uses solve_ivp from the SciPy package in Python, and the results from the simulation are shown in Figure 3. Initially, A_β and Ca increase and drive each other's growth. This leads to a large growth of τ proteins, thus increasing neuron loss over time. The biomarker populations

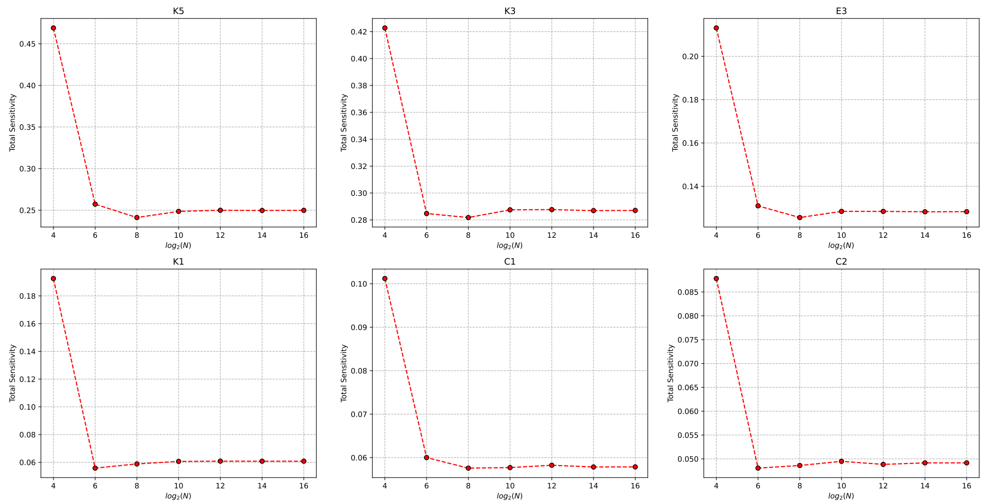

Figure 2: 6 most sensitive parameters for $N \in [2^4, 2^{16}]$

reach their equilibrium values relatively early, but it takes cognitive decline many years to reach its equilibrium value.

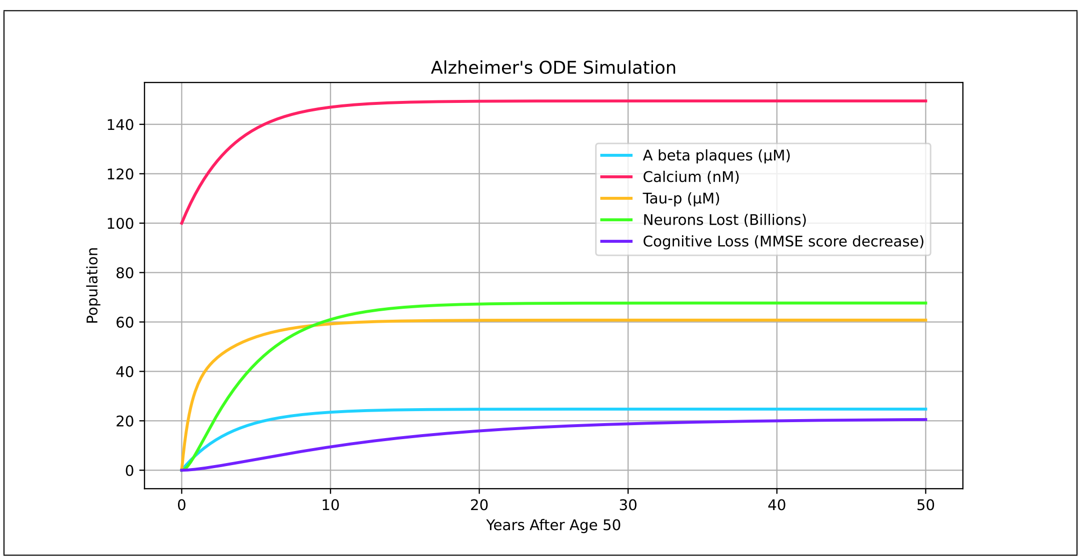

Figure 3: ODE system simulation

A phase portrait, shown in Figure 4, displays how different trajectories with varying initial conditions traverse the $A_\beta - Ca$ space. Trajectories will quickly move toward the Ca nullcline with slight movement along the horizontal axis. This is because the time-scale of Ca is much smaller than that of A_β. Once trajectories are nearby the nullcline, they along the nullcline toward the equilibrium solution.

To determine which proteins are most effective to treat, we graph the equilibrium cognitive decline value against varying treatments against each protein, as seen in Figure 5. Each

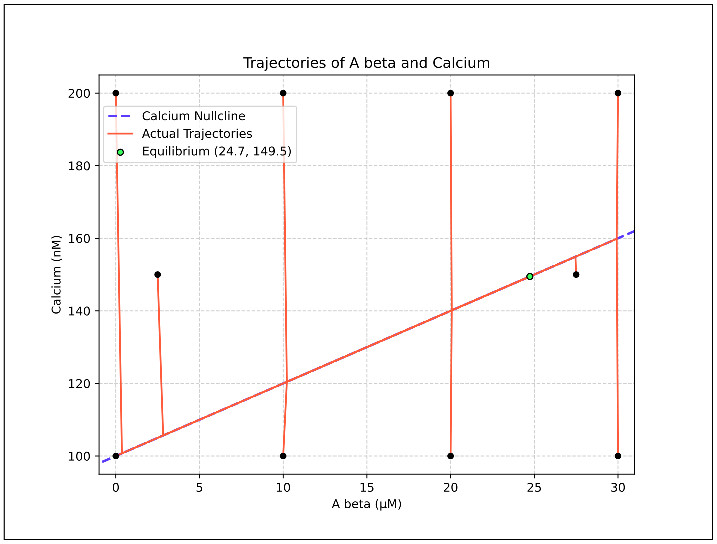

Figure 4: Phase portrait

u_n term increases such that the behavior of the function can be observed. We see that the variable that has the largest impact on C^* is u_3, the treatment that reduces the number of τ proteins. The next most effective treatment is u_1, which reduces the concentration of A_β plaques in the brain. We observe that treating the average level of Ca does not have a large effect on the equilibrium cognitive decline.

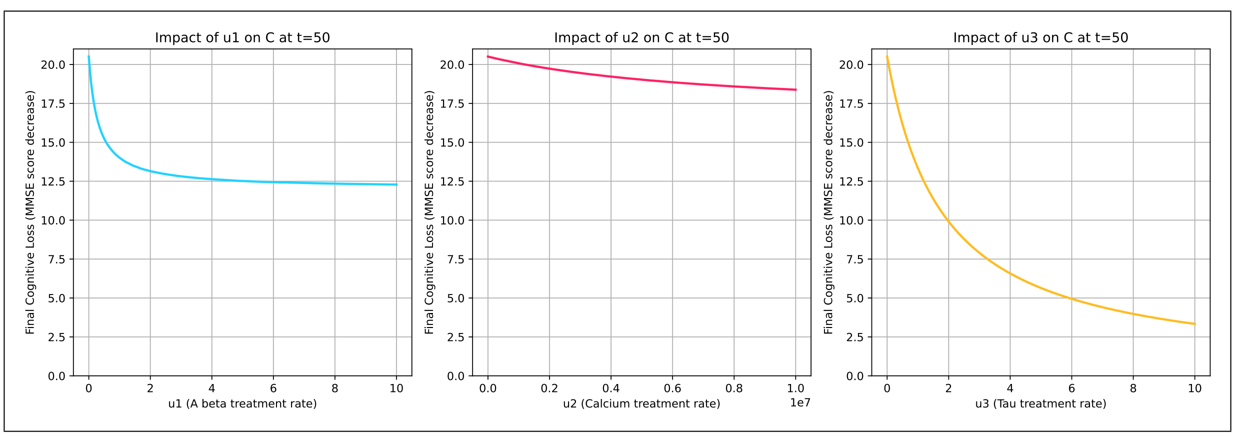

Figure 5: Treatment effect on C^*

7 Optimal Control

We use optimal control to determine the optimal values of u_n to minimize the total cost of treatments and results. In optimal control, we will let each u_n be a function of time, $u_n(t)$, as it is not realistic to maintain a constant treatment long-term. Tracking the cost of treatments is useful for our application as some drugs used to treat Alzheimer's Disease have been shown to induce strong side effects such as brain swelling and brain bleeding [5]. So, minimizing this cost function while maximizing treatments will let the medication have its strongest impact without adverse effects.

7.1 Objective Function

The following equation is our objective function which tracks our cost over the time horizon of the treatment.

$$J(u_1, u_2, u_3) = \int_0^{t_f} A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 \, dt. \tag{7}$$

The goal of treatment is to minimize the equilibrium values N^* and C^* while at the same time minimizing the side effects from the drugs. The terms A_n are the weights which determine how important each associated variable is in the optimization problem. A_3, A_4, and A_5 are the weights associated with the quadratic cost of treatments u_n that target A_β, calcium, and τ protein levels. Also, t_f is the final time of the control. The set of admissible control functions is given by:

$$u = \left\{(u_1, u_2, u_3) \in L^\infty(0, t_f)^3 \mid (u_1, u_2, u_3) \in [0, u_1^{\max}], [0, u_2^{\max}], [0, u_3^{\max}] \; \forall t \in [0, t_f]\right\}$$

7.2 Hamiltonian

$$\begin{aligned}\mathcal{H} = {} & A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 + \lambda_1 \left(a_1 + a_2 \frac{Ca}{Ca + \sigma_1} - k_1 A_\beta - u_1 A_\beta\right) \\ & + \lambda_2 \left(b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca\right) + \lambda_3 \left(c_1 + c_2 A_\beta + c_3 \frac{Ca}{Ca + \sigma_2} - k_3 \tau - u_3 \tau\right) \\ & + \lambda_4 \left(d_1 + d_2 \tau - k_4 N\right) + \lambda_5 \left(e_1 + e_2 N R + e_3 \tau - k_5 C\right).\end{aligned}$$

The Hamiltonian includes the treatment costs as described in Equation (7) as well as the co-state vectors for each variable present in our system. Thus, we seek an optimal control set (u_1^*, u_2^*, u_3^*) such that

$$J(u_1^*, u_2^*, u_3^*) = \min\left\{J(u_1, u_2, u_3) \mid (u_1, u_2, u_3) \in \mathcal{U}\right\}.$$

Theorem 2 (Existence of Optimal Control)**.** *There exists an optimal control set* (u_1^*, u_2^*, u_3^*) *that minimizes* $J(u_1, u_2, u_3)$ *over* $\mathcal{U}$*. Furthermore, there exists adjoint variables* $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ *and* λ_5 *that satisfy the following system of differential equations:*

$$
\begin{aligned}
\frac{d\lambda_1}{dt} &= -\frac{\partial \mathcal{H}}{\partial A_\beta} = (k_1+u_1)\lambda_1 - b_2\lambda_2 - c_2\lambda_3 \\
\frac{d\lambda_2}{dt} &= -\frac{\partial \mathcal{H}}{\partial Ca} = -\frac{a_2\sigma_1\lambda_1}{(Ca+\sigma_1)^2} + (k_2+u_2)\lambda_2 - \frac{c_3\sigma_2\lambda_3}{(Ca+\sigma_2)^2} \\
\frac{d\lambda_3}{dt} &= -\frac{\partial \mathcal{H}}{\partial \tau} = (k_3+u_3)\lambda_3 - d_2\lambda_4 - e_3\lambda_5 \\
\frac{d\lambda_4}{dt} &= -\frac{\partial \mathcal{H}}{\partial N} = -A_2 + k_4\lambda_4 - e_2R\lambda_5 \\
\frac{d\lambda_5}{dt} &= -\frac{\partial \mathcal{H}}{\partial C} = -A_1 + k_5\lambda_5
\end{aligned}
\tag{8}
$$

with transversality conditions $\lambda_1(t_f) = \lambda_2(t_f) = \lambda_3(t_f) = \lambda_4(t_f) = \lambda_5(t_f) = 0$. *Also, the optimal control set has the characterization:*

$$
\begin{aligned}
u_1^* &= \min\left\{u_1^{\max}, \max\left\{0, \frac{\lambda_1 A_\beta}{2A_3}\right\}\right\} \\
u_2^* &= \min\left\{u_2^{\max}, \max\left\{0, \frac{\lambda_2 Ca}{2A_4}\right\}\right\} \\
u_3^* &= \min\left\{u_3^{\max}, \max\left\{0, \frac{\lambda_3 \tau}{2A_5}\right\}\right\}.
\end{aligned}
$$

Proof. Using Corollary 4.1 of [6], the existence of an optimal control set follows from the Lipschitz property of system (4) with respect to the state variables, *a priori* boundedness of the state solutions established in Subsection 3.1, and the convexity of the integrand of J with respect to (u_1, u_2, u_3).

By applying Pontryagin's Maximum Principle, the adjoint variables must satisfy system (8) with zero final time conditions. To obtain the characterizations of the optimal controls, we solve $\frac{\partial \mathcal{H}}{\partial u_1} = \frac{\partial \mathcal{H}}{\partial u_2} = \frac{\partial \mathcal{H}}{\partial u_3} = 0$ on the interior of the control set $\mathcal{U}$. Using the bounds on the controls, we obtain the desired characterization. □

7.3 Numerical Results

- explain algorithm here

Weights are calculated by initially running the ODE simulation with each $u_n(t) = u_n^{\max}/2$. We choose $A_1 = 1$ using the extra degree of freedom. Using the fact that

$$
A_1 N_{\text{avg}} = A_2 C_{\text{avg}} = \int_0^{t_f} u_1^2\, dt = \int_0^{t_f} u_2^2\, dt = \int_0^{t_f} u_3^2\, dt,
$$

we use $A_1 = 1 \implies A_1 N_{\text{avg}} = N_{\text{avg}}$ to find that $A_2 = \frac{N_{\text{avg}}}{C_{\text{avg}}}$. Notice that each u_n is constant, so $\int_0^{t_f} u_n^2\, dt = u_n^2 t_f$, thus $A_3 = \frac{N_{\text{avg}}}{u_1^2 t_f}$, $A_4 = \frac{N_{\text{avg}}}{u_2^2 t_f}$, and $A_5 = \frac{N_{\text{avg}}}{u_3^2 t_f}$. The optimality system

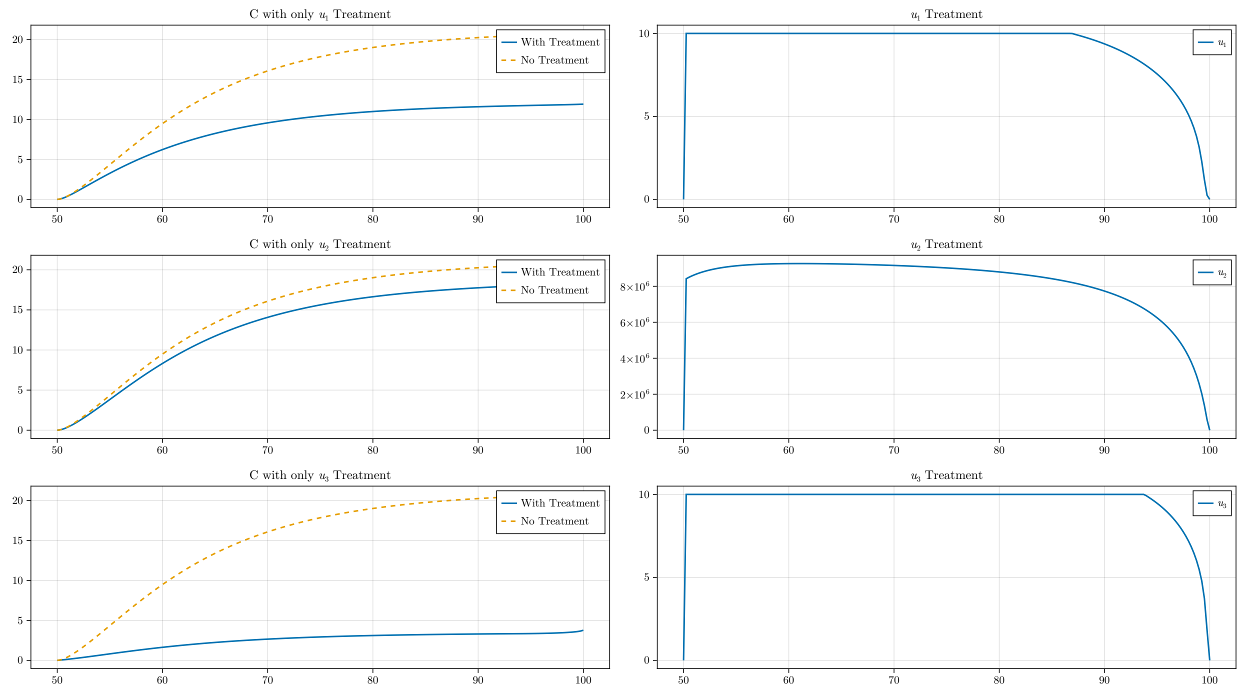

Figure 6: Optimal control simulation results

was solved using parameter values found in Table 1, a final time of $t = 50$, and the initial conditions $(A_\beta, Ca, \tau, N, C) = (0, 100, 0, 0, 0)$.

Figure 6 shows the optimal control profile with the values of A_β, calcium, and tau proteins over time, with and without treatments. The value of the function $u_n(t)$ is shown over the duration of the treatment. After running numerical simulations for optimal control, we find that the results are in line with the simulation shown in Figure 5. That is, treatments that target tau protein are most effective in reducing C^*, followed by treatments targeting A_β. Again, treatments targeting Ca are found to be the least effective in reducing C^*. In each case, the treatment is maximally applied for around 40 years, and then tapers off for the remaining 10 years.

8 Conclusion

In this paper we developed an ODE model for the progression of Alzheimer's Disease as a result of increasing populations of beta-amyloid proteins, intracellular calcium dysregulation, and tau proteins, which then led to neuron loss and cognitive decline. We completed a global stability analysis in the applicable region and then performed sensitivity analysis on the parameters of the model. The model is found to have one globally asymptotically stable equilibrium solution which changes as treatments are applied to the patient. Constant treatment projections were used to make an early hypothesis on the efficacy of targeting different biomarkers.

We used a Runge-Kutta Forward Backward sweep to determine optimal control inputs for the model in order to minimize the cost of treating a patient. The results of the optimal control analysis showed that it is most effective to target tau protein populations. Including intracellular calcium in our model shows that beta-amyloid is not the only upstream influence on the development Alzheimer's Disease, thus giving explanation as to why beta-amyloid medications have been ineffective in clinical trials. Optimal treatments on tau proteins results in a significant decrease in cognitive decline as compared to beta-amyloid and calcium treatments.

The model presented has limitations which provide different avenues for further research. To have more tractable solutions, the model uses Hill equations of $n = 1$, although higher order equations more accurately describe the biological relationships between biomarkers. The model can be improved by using Alzheimer's Disease data sets to hone in on biologically accurate parameters.

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